Reg. No. | | | | | | | | | | [18CS5MA01]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

MOBILE APPLICATION DEVELOPMENT USING OPEN SOURCE

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Explain Android Activity Life Cycle with a neat diagram. **07**

b. Compare RelativeLayout with LinearLayout and design an example UI and apply Linearlayout and RelativeLayout. **07**

2. a. With an example, show how click event handler can be handled through static and dynamic way. **06**

b. Write short notes on the following UI elements in Android with example **08**

i. ListView
ii. RadioButton
iii. Alert Dialogs
iv. CheckBox

3. a. Android provides many kinds of data storage for applications. Compare the different ways of storing data in android through appropriate examples. **10**

b. List out and explain the advantages of SQLite databases. **04**

OR

4. a. Create a SQLite database and perform CRUD (create, read, update and delete) operations on it and explain with a sample App Development codes in Java and XML. **10**

b. Write the best storage mechanism for the following scenarios: **04**

i. To fetch the last software updated date
ii. Minimal traffic and structured data
iii. Application specific files
iv. Share data across applications

5. a. Explain the term Notification and its purpose in Android. With the help of an example write down the steps to create and send a notification. **07**

b. With the help of a diagram explain Android Services life cycles and different types of services. **07**

OR

6.	**a.**	Build an app that delivers a notification when the time is 02:00 PM. (using Java)	**07**
	b.	Write a note on Android Sensors in detail.	**07**
7.	**a.**	What are mutable and immutable variables in Kotlin and how does the compiler determine the type of these variables? Give examples.	**07**
	b.	Illustrate Primary constructor and Secondary constructor through appropriate code snippets.	**07**

OR

8.	**a.**	Differentiate in between Unit Test and Instrumentation Test.	**07**
	b.	Discuss Kotlin flow control statements with examples.	**07**